

Education

Nanjing University

Sep 2023 – present

B.Eng. in Intelligent Software Engineering (expected Jun 2027)

School of Intelligent Software and Engineering

Average: 88.4/100

Research Interests

I study how AI systems can learn and improve continually from data, experience, and human feedback, with an emphasis on:

- **Data-centric AI:** diagnosing model failures and guiding targeted data improvements through interpretable evaluation and visual analytics.
- **Self-improving agents:** learning reusable skills and memory from experience, validating behavioral changes, and transferring learned capabilities across tasks and models.
- **Human-AI interaction:** designing interfaces that help people inspect, evaluate, and steer model behavior and agent learning.

Research Experience

The Hong Kong University of Science and Technology (Guangzhou)

Visiting Research Student

Aug – Oct 2026

Research with Prof. Weikai Yang

Nanjing University, iMATE Lab

Mar 2025 – present

Research Assistant | Advisor: Prof. Ke Xu

School of Intelligence Science and Technology; collaboration with Dr. Wenchao Li.

Publications and Preprints

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations

Wenchao Li, Yuewen Gao, Yu He, Cong Zhu, Ke Xu.

CHI 2026 | Published conference paper | DOI

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion

Yu He, Weikai Yang.

Submitted to AAI 2027 | arXiv preprint, 2026 | Paper

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning

Yiyang Lu, Yu He, Jianlong Chen, Hongyuan Zha.

arXiv preprint, 2026 | Paper

Manuscripts Under Review

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation

Submitted to CHI 2027

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence

Submitted to CHI 2027

Selected Research Projects

SliceTune First author; research lead

- Designed and built a visual analytics system that maps validation goals to data attributes through a Feature Graph, letting practitioners steer failure-slice discovery.
- Led a comprehensive evaluation combining case studies in image, audio, and text classification, a controlled user study, and expert interviews to assess failure discovery, diagnostic support, and data-guided model improvement.

SkillContour First author; research lead

- Designed and built a workbench that links source patches, skill instructions, and execution evidence to diagnose faulty consolidation and support regrouping, re-synthesis, and replay.
- Combined a controlled user study with evaluation of rule-based contract reasoning on unseen tasks. The system strengthened perceived support for skill diagnosis and improved agent task accuracy over automatic consolidation and text-based review.

SkillCommit First author; research lead

- Designed and implemented skill consolidation using cross-instance replay and mechanism checks to assess compatibility, validating merged skills on source cases before commitment.
- Demonstrated gains in complex rule application, legal reasoning, and abstract logical deduction and cipher decoding, without parameter updates; learned skills also transferred across model sizes and families.

NetworkCanvas Mar – Sep 2025

- Implemented visualization and feedback components for network exploration, contributing to the data integration and interactions behind adaptive recommendations and branching histories.
- Contributed to figures and study documentation; controlled studies found more noteworthy observations and higher analytical confidence than exploration without recommendations.

MSSR Aug 2025 – Jan 2026

- Modeled knowledge retention and implemented a replay scheduler that uses sample-level memory estimates to decide when to rehearse earlier examples.
- Evaluated continual fine-tuning across task sequences to assess knowledge retention and adaptation to new tasks. Adaptive replay improved performance over compared replay baselines, with particularly strong gains in reasoning.

Selected Honors and Awards

Nanjing University People’s Scholarship Nov 2025

Meritorious Winner, Mathematical Contest in Modeling
Consortium for Mathematics and Its Applications (COMAP). May 2025

National Third Prize, China Collegiate Innovation and Invention Competition Dec 2024

Technical Skills

- Programming:** Python, TypeScript, JavaScript, Java, C++, SQL.
- Machine learning:** PyTorch; continual fine-tuning and agent evaluation.
- Interactive systems:** React, Vue 3, D3.js, G6.js, ECharts; Git/GitHub.
- Languages:** Chinese (native); English.